

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is a common scenario where transfer learning is highly beneficial?

1 / 1 point

- ☐ Creating a model for data entirely unrelated to existing datasets
- ☐ When perfect classification is required from the onset
- ☐ Developing an app without concerning learning algorithms
- ☒ Working with a small, domain-specific dataset similar to a large annotated dataset

✔ Correct
Correct! This is a typical use case for transfer learning, leveraging a large dataset's features.

2. Which optimizer is known for combining adaptive learning rate and momentum descent?

1 / 1 point

- ☐ SGD (Stochastic Gradient Descent)
- ☐ Nesterov Accelerated Gradient
- ☐ RMSProp
- ☒ Adam W

✔ Correct
Correct! Adam W features adaptive learning rates and momentum, suitable for complex models.

3. Why is specifying a batch size important in machine learning model training?

1 / 1 point

- ☐ It determines the total number of epochs
- ☐ It ensures models aren't overfitting
- ☐ It dictates the final model accuracy
- ☒ It balances memory usage and training speed

✔ Correct
Correct! Batch size is crucial for efficient memory use while maintaining a practical training pace.

4. Why might object detection models require multiple output heads?

1 / 1 point

- ☐ To speed up the processing time of each image
- ☐ To account for multiple resolutions of input images
- ☐ To provide backup in case one head fails
- ☒ To predict both class and location of multiple objects

✔ Correct
Correct! This enables the model to provide detailed outputs on object type and position.

5. What does resizing images in a machine learning model help to achieve?

1 / 1 point

- ☒ Standardizes input dimensions for consistent model processing
- ☐ Increases the amount of training data
- ☐ Enhances classification accuracy automatically
- ☐ Improves the visual quality of images

✔ Correct
Correct! This is crucial for ensuring model input expectations are met uniformly.